# What Will Students Share? Consent and Comfort with Fine-Grained Telemetry in Educational IDEs

Colm Brennan, Wei Zhou, Ruth Adler

arXiv:2602.11873 [cs.CY], 17 Feb 2026

## Abstract

Research on novice programming increasingly depends on fine-grained telemetry, yet little is known about what students are willing to share. We surveyed 418 students at three universities about eight data streams, from submission history to webcam video, and four uses, from aggregate research to individual instructor review. Comfort was high for submissions (89%) and keystrokes (61%) but low for assistant prompts (38%), which students saw as revealing confusion. Restricting use to aggregate dashboards raised comfort by 22 percentage points on average. We propose a consent design that separates streams and uses.

# 1 Introduction

Learning-analytics consent forms often ask for a single yes or no to "data collected by the IDE". Students may feel very differently about a log of runs and a log of their questions to an assistant. We measure these differences.

# 2 Survey

Participants (N = 418; 3 universities; CS1 and CS2) rated comfort on a 5-point scale for each combination of eight streams and four uses. We counted ratings of 4 or 5 as comfortable. The survey also included two open questions about what they would want to know before agreeing.

# 3 Results

| Stream | Comfortable (any use) | Aggregate only | Instructor reviews my session |
|---|---|---|---|
| Submission history | 89% | 94% | 81% |
| Run and test results | 78% | 90% | 66% |
| Keystrokes | 61% | 82% | 40% |
| Assistant prompts | 38% | 63% | 19% |
| Screen recording | 21% | 37% | 9% |
| Webcam | 6% | 11% | 3% |

Assistant prompts were the least comfortable text stream. In open responses, 112 students said prompts showed "what I don't know" or were "like a diary of being confused". Several students were more comfortable with an automated classifier reading prompts than with a person reading them, provided only labels were stored.

# 4 Consent design

We recommend (1) separate consent per stream; (2) separate consent for aggregate research and individual review; (3) storing derived labels instead of raw prompts where research questions allow; (4) a retention date stated in the form; and (5) a dashboard where students can see what was recorded

about them.

# 5 Limitations

Hypothetical comfort is not behavior; actual opt-out rates in deployments are lower than these numbers suggest.

# References

[1] Haldane, I., Okafor, T., and Brandt, L. Replaying the keystroke. SIGCSE TS 2024.

[2] Vasquez, D., Ahmed, K., and Petrov, I. Where did this code come from? Koli Calling 2024.